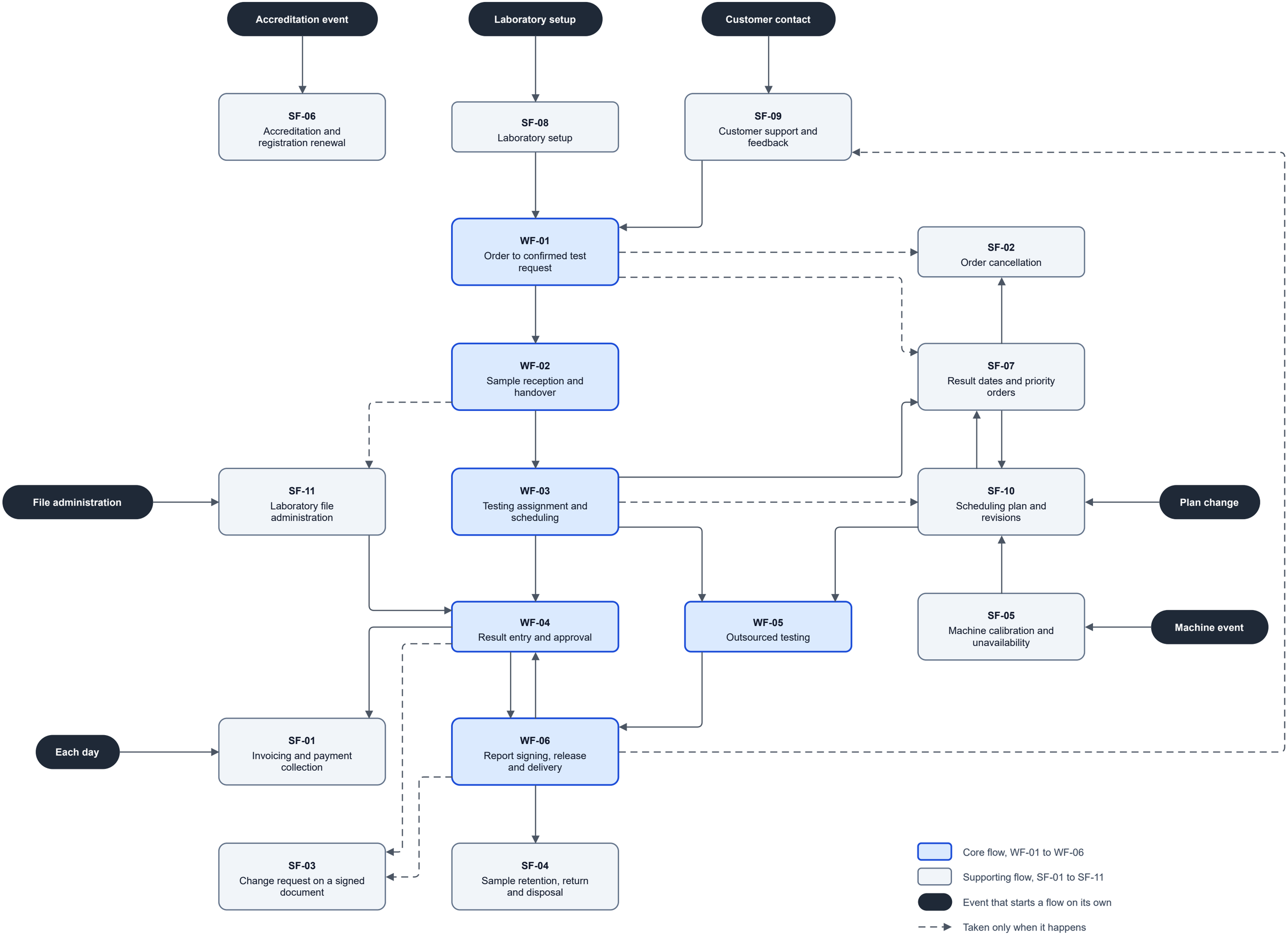

Accreditation event
Laboratory setup
Customer contact
SF-06
Accreditation and registration renewal
SF-08
Laboratory setup
SF-09
Customer support and feedback
WF-01
Order to confirmed test request
SF-02
Order cancellation
WF-02
Sample reception and handover
SF-07
Result dates and priority orders
File administration
SF-11
Laboratory file administration
WF-03
Testing assignment and scheduling
SF-10
Scheduling plan and revisions
Plan change
WF-04
Result entry and approval
WF-05
Outsourced testing
SF-05
Machine calibration and unavailability
Machine event
Each day
SF-01
Invoicing and payment collection
WF-06
Report signing, release and delivery
SF-03
Change request on a signed document
SF-04
Sample retention, return and disposal
Core flow, WF-01 to WF-06
Supporting flow, SF-01 to SF-11
Event that starts a flow on its own
Taken only when it happens

WF-01 Order to confirmed test request

Customer	Sales	System	Head of Sales	Accountant

Customer opens portal

Fill cart

Submit order

the customer wants to cancel

→ SF-02 Order cancellation

requested date or priority

→ SF-07 Result dates and priority orders

Queue order

Claim order

Confirm ways and expected samples

estimate the return date

→ SF-07 Result dates and priority orders

Confirm price and percentage discount

Above quotation approval thresholds?

yes

Request approval

Notify Head of Sales

Review quotation

Approve?

no

Notify sales

Adjust quotation

no

yes

Sign approval

Send quotation

Post advance invoice

Email quotation

Accept quotation?

no

Decline with reason

Notify sales

yes

Revise quotation

Send new version

Lock old version

Pay online?

no

Record transfer or cash payment

Confirm payment

yes

Pay advance (PayOS)

Record payment

Accept quotation

Confirm order

Create test request

Pay balance (PayOS)

no

Paid or on credit?

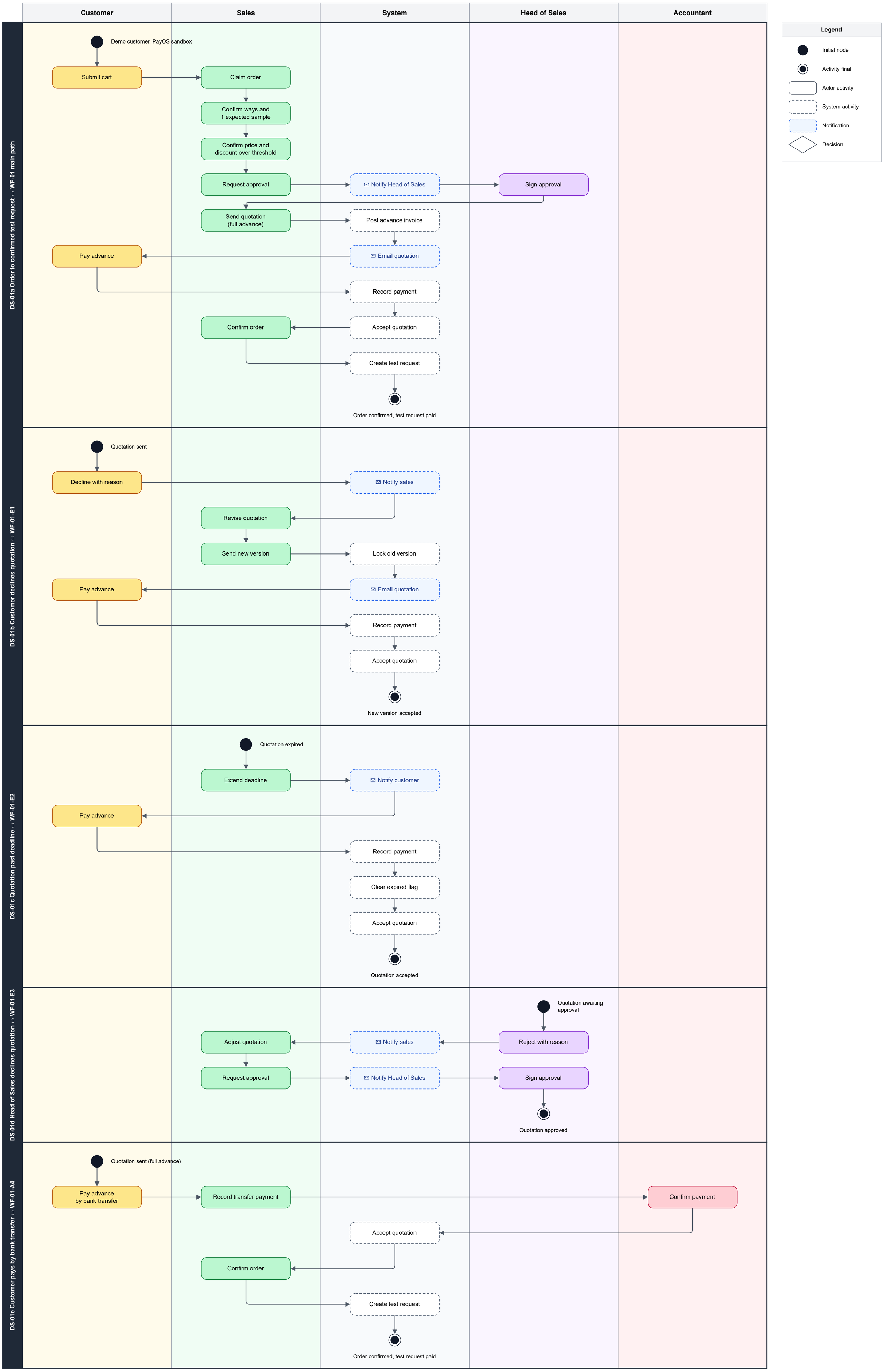
Record payment

yes

Test request paid → WF-02 Sample reception and handover

Legend

- Initial node
- Activity final
- Flow final
- Actor activity
- System activity
- Notification
- Decision


Customer
Sales
System
Head of Sales
Accountant
Legend
Initial node
Activity final
Actor activity
System activity
Notification
Decision
DS-01a Order to confirmed test request ↔ WF-01 main path
Demo customer, PayOS sandbox
Submit cart
Claim order
Confirm ways and 1 expected sample
Confirm price and discount over threshold
Request approval
Notify Head of Sales
Sign approval
Send quotation (full advance)
Post advance invoice
Pay advance
Email quotation
Record payment
Confirm order
Accept quotation
Create test request
Order confirmed, test request paid
DS-01b Customer declines quotation ↔ WF-01-E1
Quotation sent
Decline with reason
Notify sales
Revise quotation
Send new version
Lock old version
Pay advance
Email quotation
Record payment
Accept quotation
New version accepted
DS-01c Quotation past deadline ↔ WF-01-E2
Quotation expired
Extend deadline
Notify customer
Pay advance
Record payment
Clear expired flag
Accept quotation
Quotation accepted
DS-01d Head of Sales declines quotation ↔ WF-01-E3
Quotation awaiting approval
Adjust quotation
Notify sales
Reject with reason
Request approval
Notify Head of Sales
Sign approval
Quotation approved
DS-01e Customer pays by bank transfer ↔ WF-01-A4
Quotation sent (full advance)
Pay advance by bank transfer
Record transfer payment
Confirm payment
Accept quotation
Confirm order
Create test request
Order confirmed, test request paid

WF-02 Sample reception and handover

Customer

Sample delivery staff

System

Head of department

Order confirmed, test request paid, samples expected
(WF-01 Order to confirmed test request)

How are the samples handed over?

collected at the customer's site

Claim the collection task and plan its time

Create collection task

Remind staff

collect again

brought to the laboratory

Collect samples

Hand over the samples

Adjust expected samples and record differences

Condition?

unusable

Reject sample

Sample rejected

good

insufficient

Put on hold

Record receipt, time, amount and storage

Notify head

Request more sample

Confirm actual samples and sign the record

Email handover PDF

Print label and hand accepted samples to departments

Scan the label to record department receipt

sample folders

extra samples or excess material

Kept with the sample, no test added; returned on request

Testing tasks in department queues
→ WF-03 Testing assignment and scheduling

→ SF-11 Laboratory file administration

→ SF-04 Sample retention, return and disposal

Legend

- Initial node
- Activity final
- Flow final
- Actor activity
- System activity
- Notification
- Decision

Customer | Sample delivery staff | System | Head of department

DS-02a Main: sample received in good condition ↔ WF-02 main path

Paid request, customer brings 1 sample

Hand over the samples

Adjust expected samples and record differences

Record receipt, time, amount and storage

Confirm actual samples and sign the record

✉ Email handover PDF

Print label and hand accepted samples to departments

Scan the label to record department receipt

Testing tasks in department queues
→ WF-03 Testing assignment and scheduling

DS-02b Sample arrives insufficient ↔ WF-02-E1

Expected sample insufficient

Record as insufficient

Put on hold

✉ Notify head

Request more sample

Claim the collection task and plan its time

Create collection task

✉ Remind staff

Collect sample

Record receipt

Sample back to received

DS-02c One of two samples unusable ↔ WF-02 unusable sample

Request with 2 samples, 1 unusable

Record as damaged

Add condition note

Reject sample

Record good sample

One rejected, one received

DS-02d Handover record has wrong quantity ↔ WF-02-E2

Signed record, wrong quantity

Create corrected record

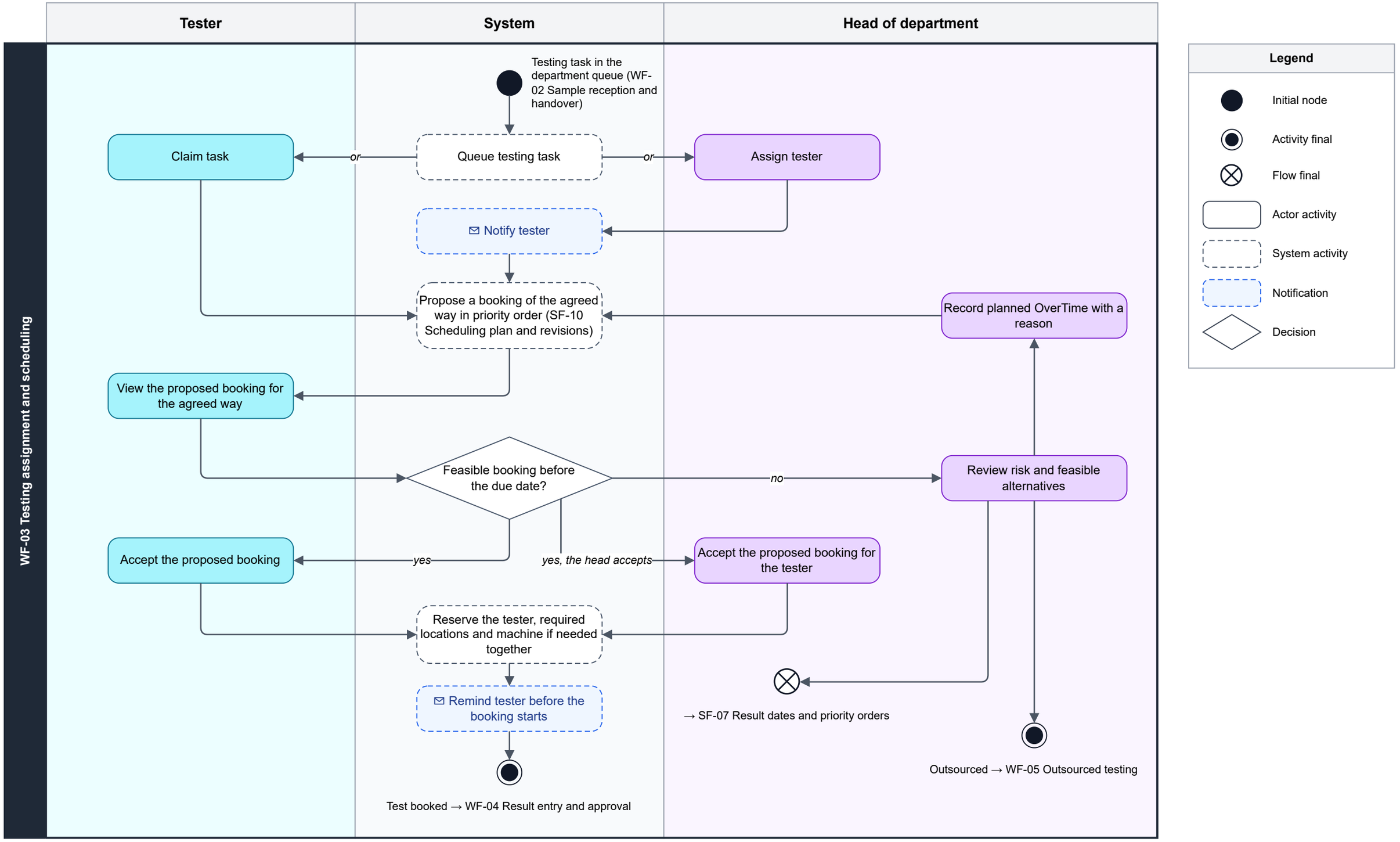
Sign corrected record

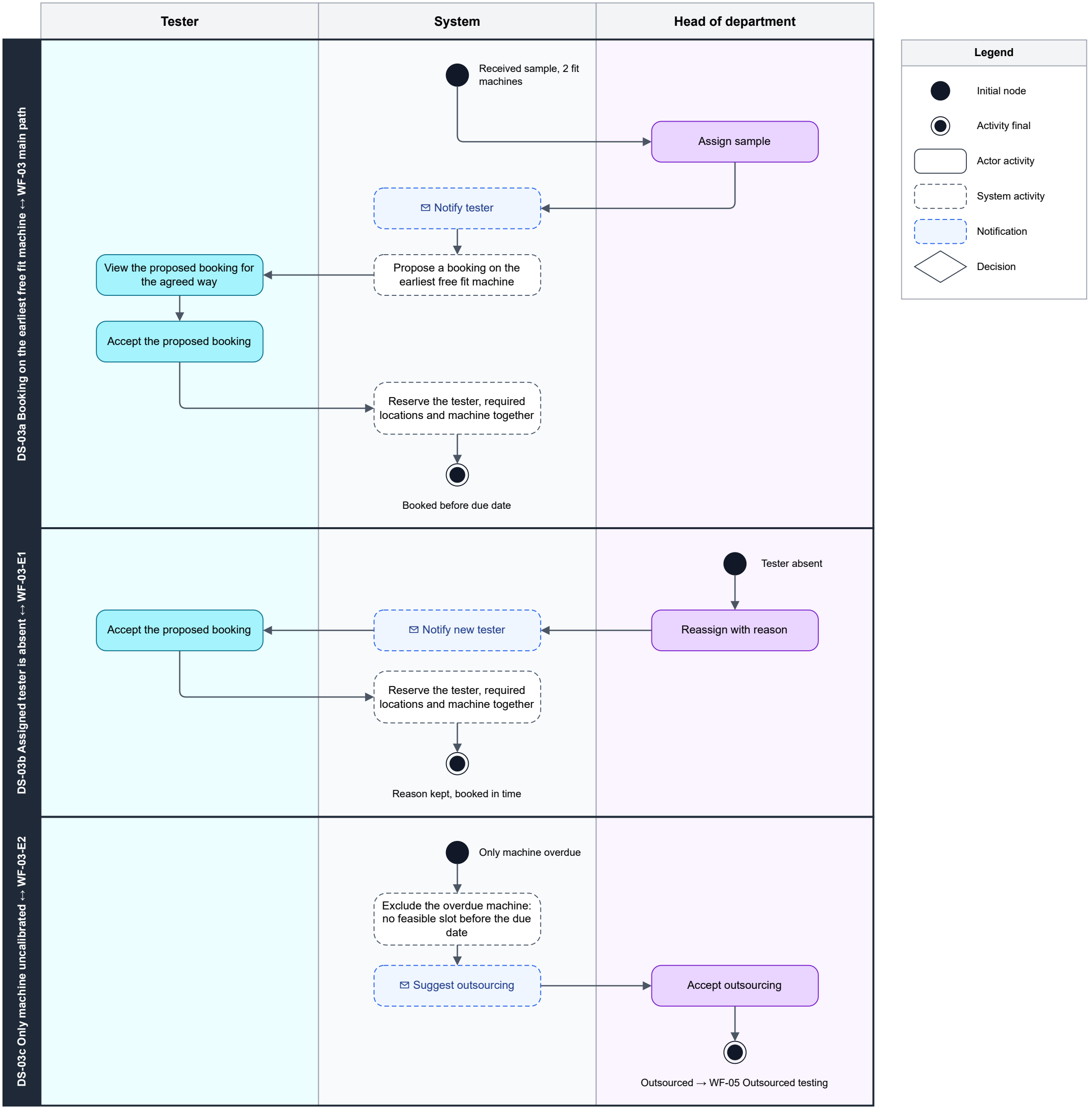
✉ Email handover PDF

New record signed, old one kept

Legend

- Initial node
- Activity final
- Actor activity
- System activity
- Notification
- Decision


WF-03 Testing assignment and scheduling
Tester
System
Head of department
Testing task in the department queue (WF-02 Sample reception and handover)
Queue testing task
or
or
Claim task
Assign tester
Notify tester
Propose a booking of the agreed way in priority order (SF-10 Scheduling plan and revisions)
Record planned OverTime with a reason
View the proposed booking for the agreed way
Feasible booking before the due date?
no
Review risk and feasible alternatives
yes
yes, the head accepts
Accept the proposed booking
Accept the proposed booking for the tester
Reserve the tester, required locations and machine if needed together
Remind tester before the booking starts
→ SF-07 Result dates and priority orders
Outsourced → WF-05 Outsourced testing
Test booked → WF-04 Result entry and approval
Legend
Initial node
Activity final
Flow final
Actor activity
System activity
Notification
Decision


Tester
System
Head of department
DS-03a Booking on the earliest free fit machine ↔ WF-03 main path
Received sample, 2 fit machines
Assign sample
Notify tester
Propose a booking on the earliest free fit machine
View the proposed booking for the agreed way
Accept the proposed booking
Reserve the tester, required locations and machine together
Booked before due date
DS-03b Assigned tester is absent ↔ WF-03-E1
Tester absent
Reassign with reason
Notify new tester
Accept the proposed booking
Reserve the tester, required locations and machine together
Reason kept, booked in time
DS-03c Only machine uncalibrated ↔ WF-03-E2
Only machine overdue
Exclude the overdue machine: no feasible slot before the due date
Suggest outsourcing
Accept outsourcing
Outsourced → WF-05 Outsourced testing
Legend
Initial node
Activity final
Actor activity
System activity
Notification
Decision

WF-04 Result entry and approval

Tester	System	Head of department

Booked test starts (WF-03 Testing assignment and scheduling); sample folders bound in Medilab Sync (SF-11 Laboratory file administration)

Follow the method and measure in the booking

Testing produces files?

yes

Save result file

Notify tester

reject with reason

Sync file version

Based on current file version?

no

Keep both conflict versions and take the server version with a local backup

no

yes

New current version

Sign the result-file version?

sign

File version signed with its hash

Download to colleagues' two-way folders

Colleagues' folders updated

Enter result with snapshot unit and LOD/LOQ

Regulation comparison requested?

yes

Copy regulation limit

no

Assess exact type limit: pass, fail or insufficient basis

Unusual result?

no

yes

Show warning; keep it on the result

Sign and send

Close testing task

Notify head

Results sent back by the Lab Head (WF-06 Report signing, release and delivery)

Review result, including own entry

Result OK?

no, reject with reason

Keep rejected result; open a new testing task

yes

Approve result

retest with a new booking

Notify tester

a mistake after completion

Every test of the sample approved or cancelled?

no

Wait for other tests

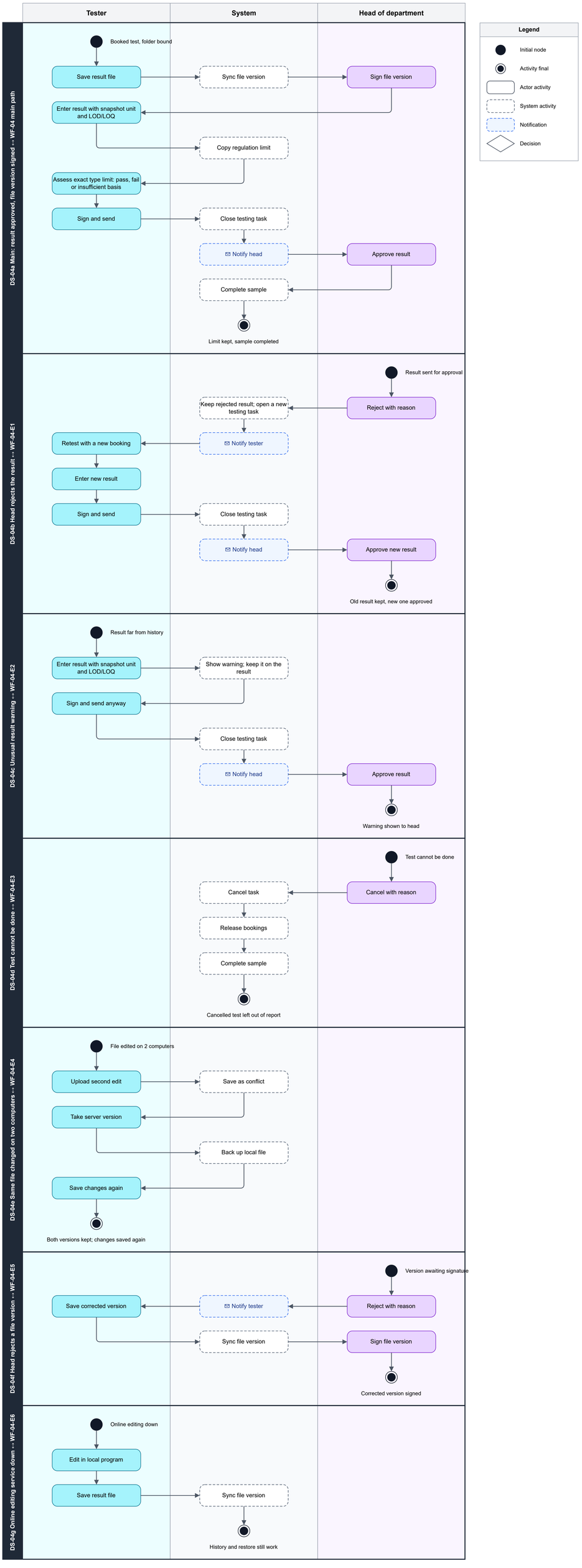
yes

Sample completed → WF-06 Report signing, release and delivery; SF-01 Invoicing and payment collection

→ SF-03 Change request on a signed document

Legend

- Initial node
- Activity final
- Flow final
- Actor activity
- System activity
- Notification
- Decision


Tester
System
Head of department
Legend
Initial node
Activity final
Actor activity
System activity
Notification
Decision
DS-04a Main: result approved, file version signed ↔ WF-04 main path
Booked test, folder bound
Save result file
Sync file version
Sign file version
Enter result with snapshot unit and LOD/LOQ
Copy regulation limit
Assess exact type limit: pass, fail or insufficient basis
Sign and send
Close testing task
Notify head
Approve result
Complete sample
Limit kept, sample completed
DS-04b Head rejects the result ↔ WF-04-E1
Result sent for approval
Keep rejected result; open a new testing task
Reject with reason
Retest with a new booking
Notify tester
Enter new result
Sign and send
Close testing task
Notify head
Approve new result
Old result kept, new one approved
DS-04c Unusual result warning ↔ WF-04-E2
Result far from history
Enter result with snapshot unit and LOD/LOQ
Show warning; keep it on the result
Sign and send anyway
Close testing task
Notify head
Approve result
Warning shown to head
DS-04d Test cannot be done ↔ WF-04-E3
Test cannot be done
Cancel task
Cancel with reason
Release bookings
Complete sample
Cancelled test left out of report
DS-04e Same file changed on two computers ↔ WF-04-E4
File edited on 2 computers
Upload second edit
Save as conflict
Take server version
Back up local file
Save changes again
Both versions kept; changes saved again
DS-04f Head rejects a file version ↔ WF-04-E5
Version awaiting signature
Save corrected version
Notify tester
Reject with reason
Sync file version
Sign file version
Corrected version signed
DS-04g Online editing service down ↔ WF-04-E6
Online editing down
Edit in local program
Save result file
Sync file version
History and restore still work

WF-05 Outsourced testing

Head of department

System

Sample delivery staff

Subcontractor

Emergency, no suggestion yet (WF-03 Testing assignment and scheduling)

No machine meets due date (WF-03 Testing assignment and scheduling)

Outsource urgently with a reason

Suggest outsourcing

Accept suggestion

Put the sample tests in a dispatch

Dispatch can go?

yes

Carry the dispatch to the subcontractor

no

Cancel dispatch

Keep tests outsourced

Record when it is sent, who took it and the expected date

Receive samples

Enter result

Notify head

Review result

Result OK?

no, reject with a reason

Keep rejected result; return test to outsourced

yes

Approve result

Notify subcontractor

Every test of the sample approved or cancelled?

no

Wait for other tests

yes

Sample completed → WF-06 Report signing, release and delivery; SF-01 Invoicing and payment collection

Legend

- Initial node
- Activity final
- Actor activity
- System activity
- Notification
- Decision

Head of department | System | Sample delivery staff | Subcontractor

DS-05a Main: subcontractor result approved ↔ WF-05 Outsourced testing, main path

- Only machine under repair
- Accept suggestion
- Put the sample tests in a dispatch
- Carry the dispatch to the subcontractor
- Record when it is sent, who took it and the expected date
- Receive samples
- Enter result
- ✉ Notify head
- Approve result
- Outsourced test approved

DS-05b Head rejects subcontractor result ↔ WF-05-E1

- Subcontractor result pending
- Reject result with a reason
- Keep rejected result; return test to outsourced
- ✉ Notify subcontractor
- Enter new result
- ✉ Notify head
- Approve result
- Test approved

DS-05c Draft dispatch cannot go out ↔ WF-05-E2

- Draft dispatch stuck
- Cancel dispatch
- Keep tests outsourced
- Put the sample tests in a new dispatch
- Carry the dispatch to the subcontractor
- Tests go with new dispatch

DS-05d Head outsources urgently ↔ WF-05-A3

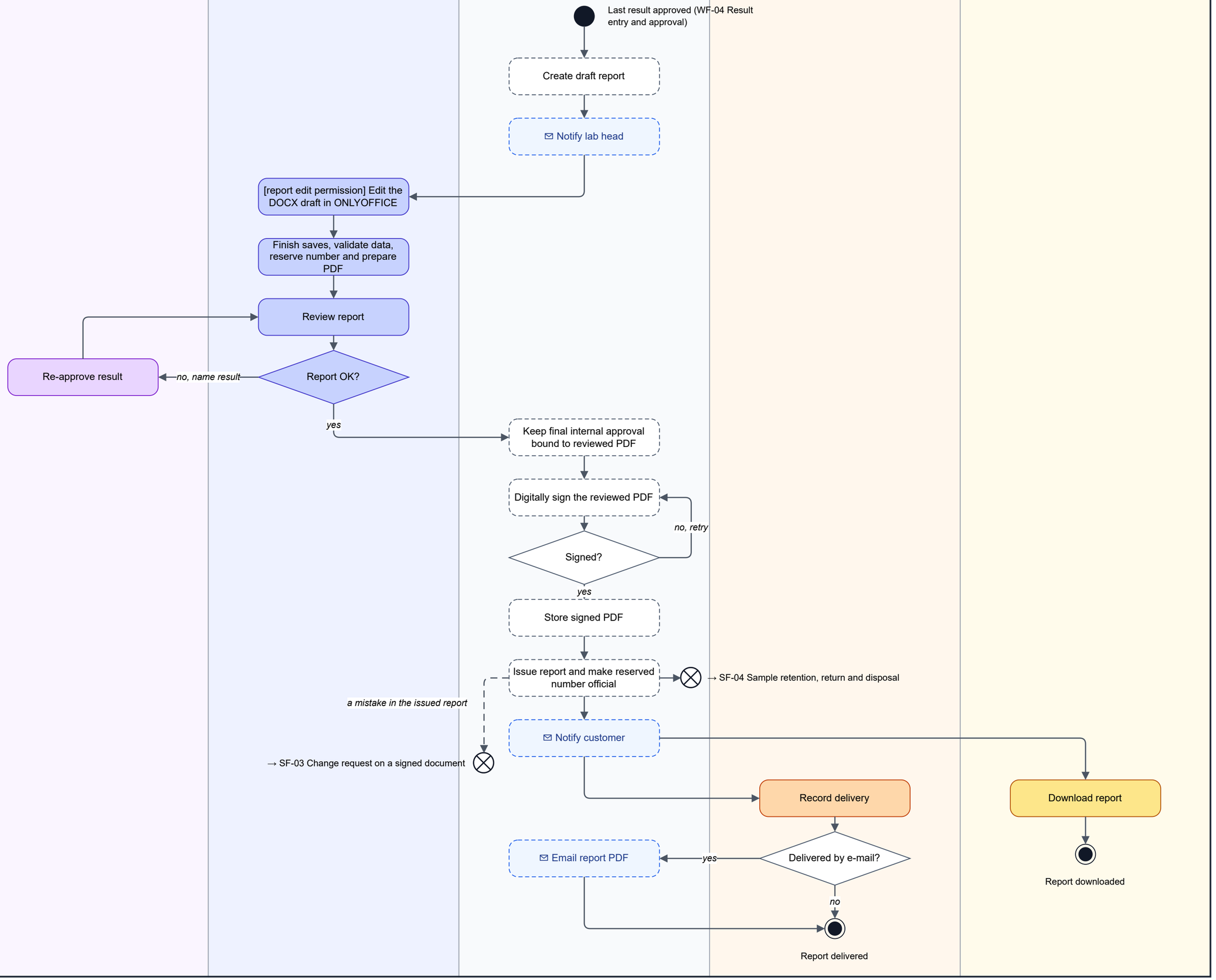
- Machine breaks; test assigned, no suggestion
- Outsource urgently with a reason
- Put the sample tests in a dispatch
- Carry the dispatch to the subcontractor
- Dispatch on its way to the subcontractor

Legend

- Initial node
- Activity final
- Actor activity
- System activity
- Notification
- Decision

WF-06 Report signing, release and delivery

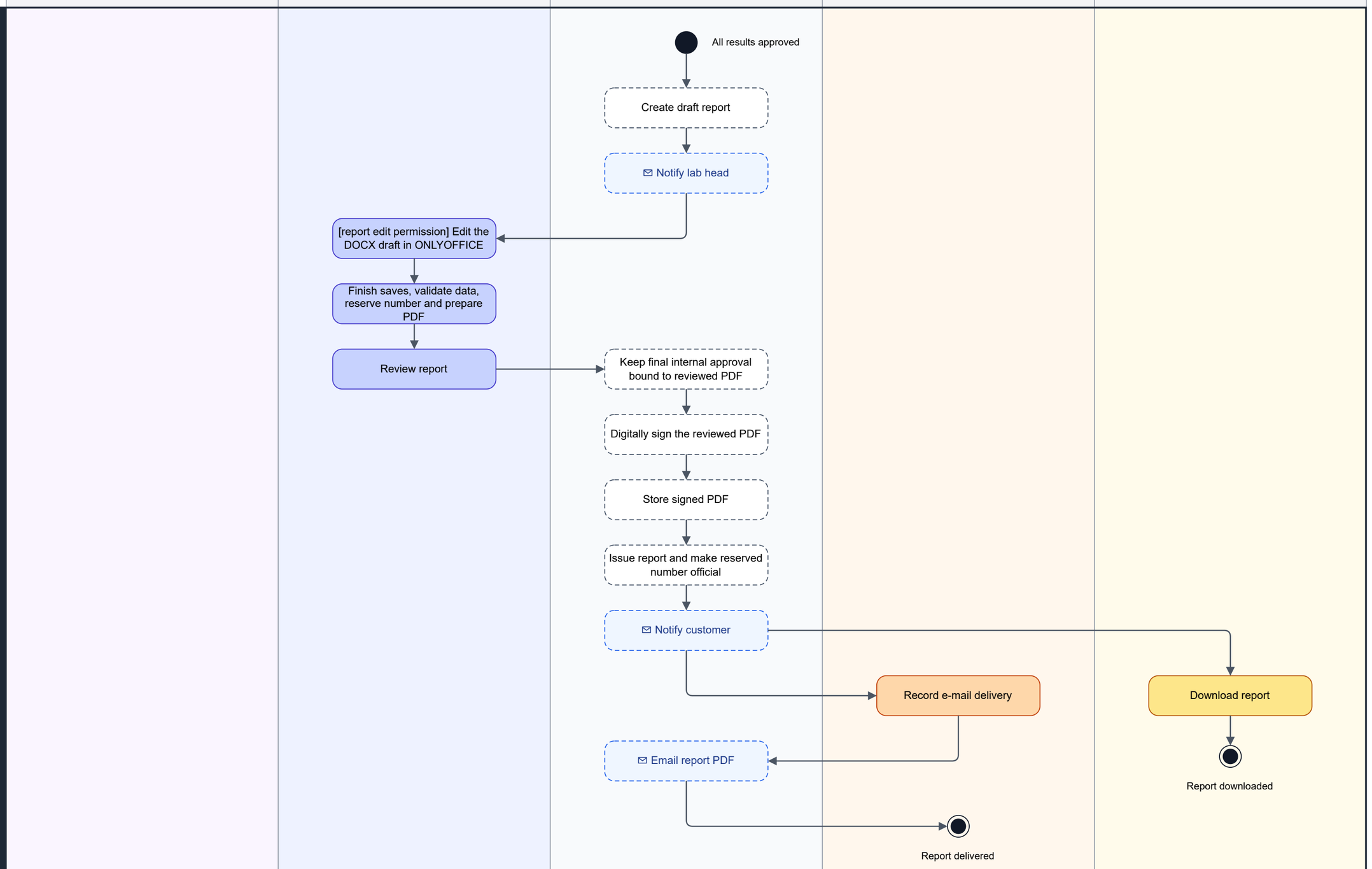
Head of department	Lab head	System	Sample delivery staff	Customer



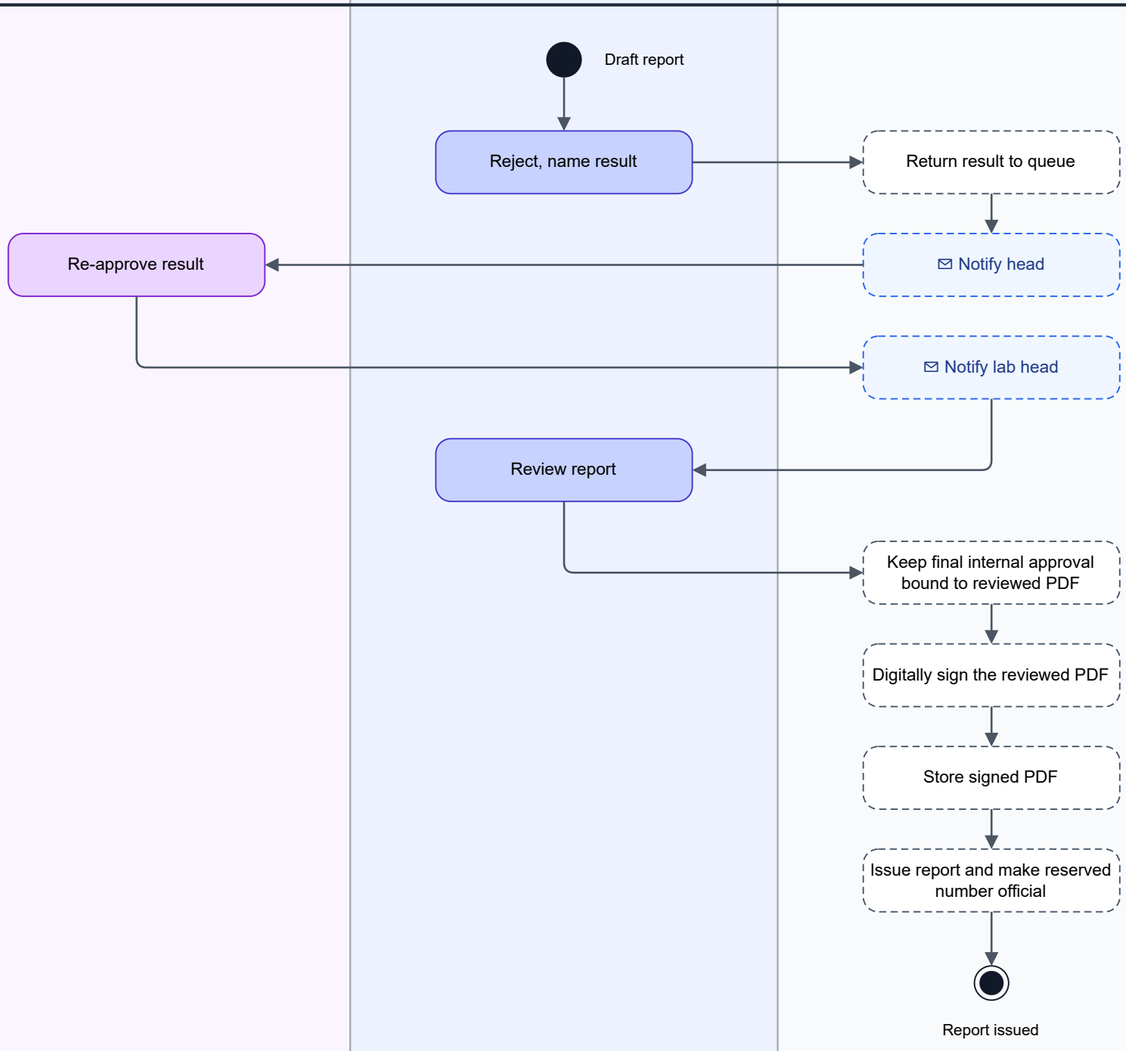
Legend

- Initial node
- Activity final
- Flow final
- Actor activity
- System activity
- Notification
- Decision

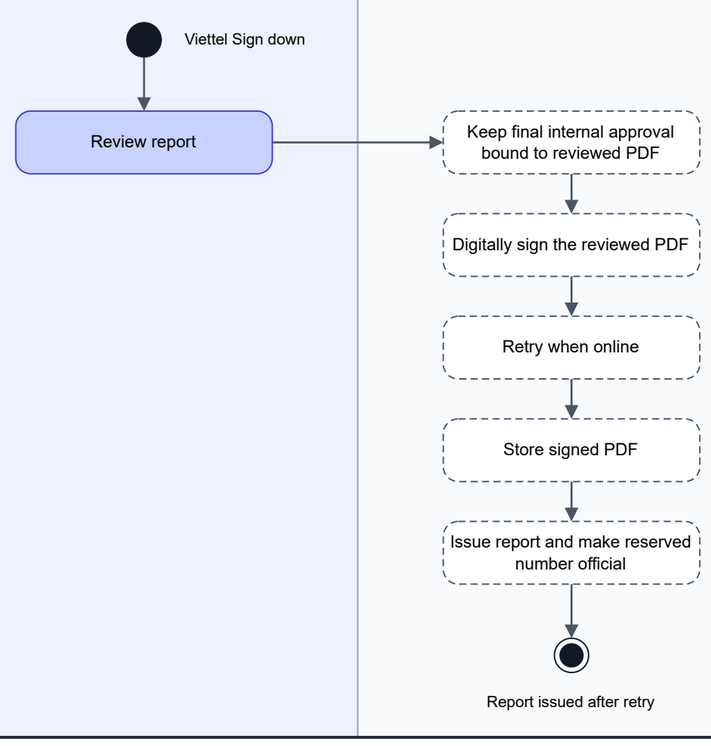
DS-06a Main: report issued and delivered ↔ WF-06 Main path

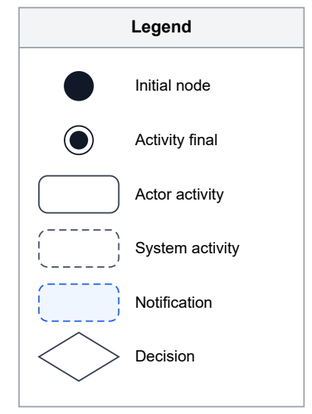


DS-06b Lab head rejects the draft report ↔ WF-06-E2

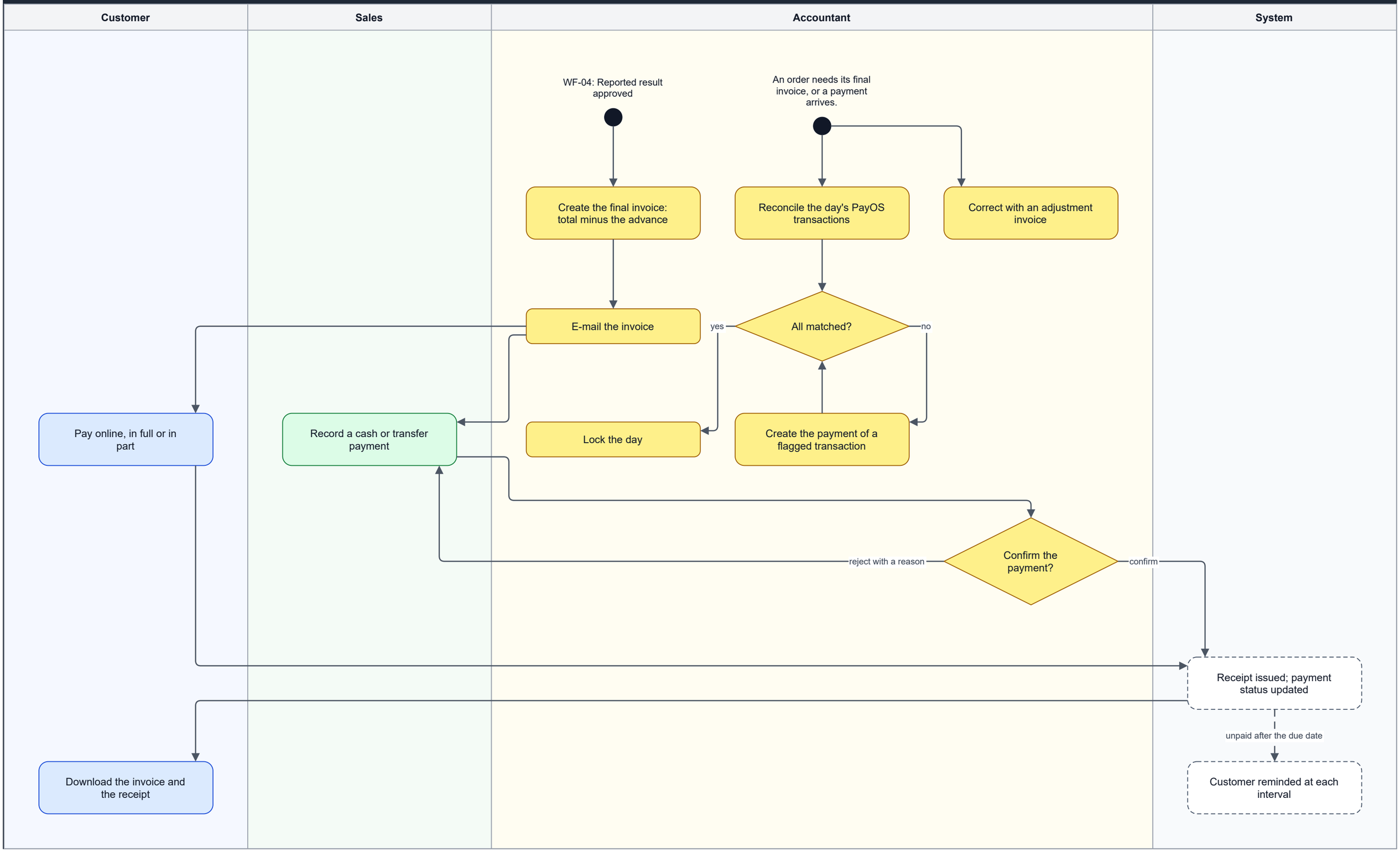


DS-06c Viettel Sign unreachable ↔ WF-06-E1

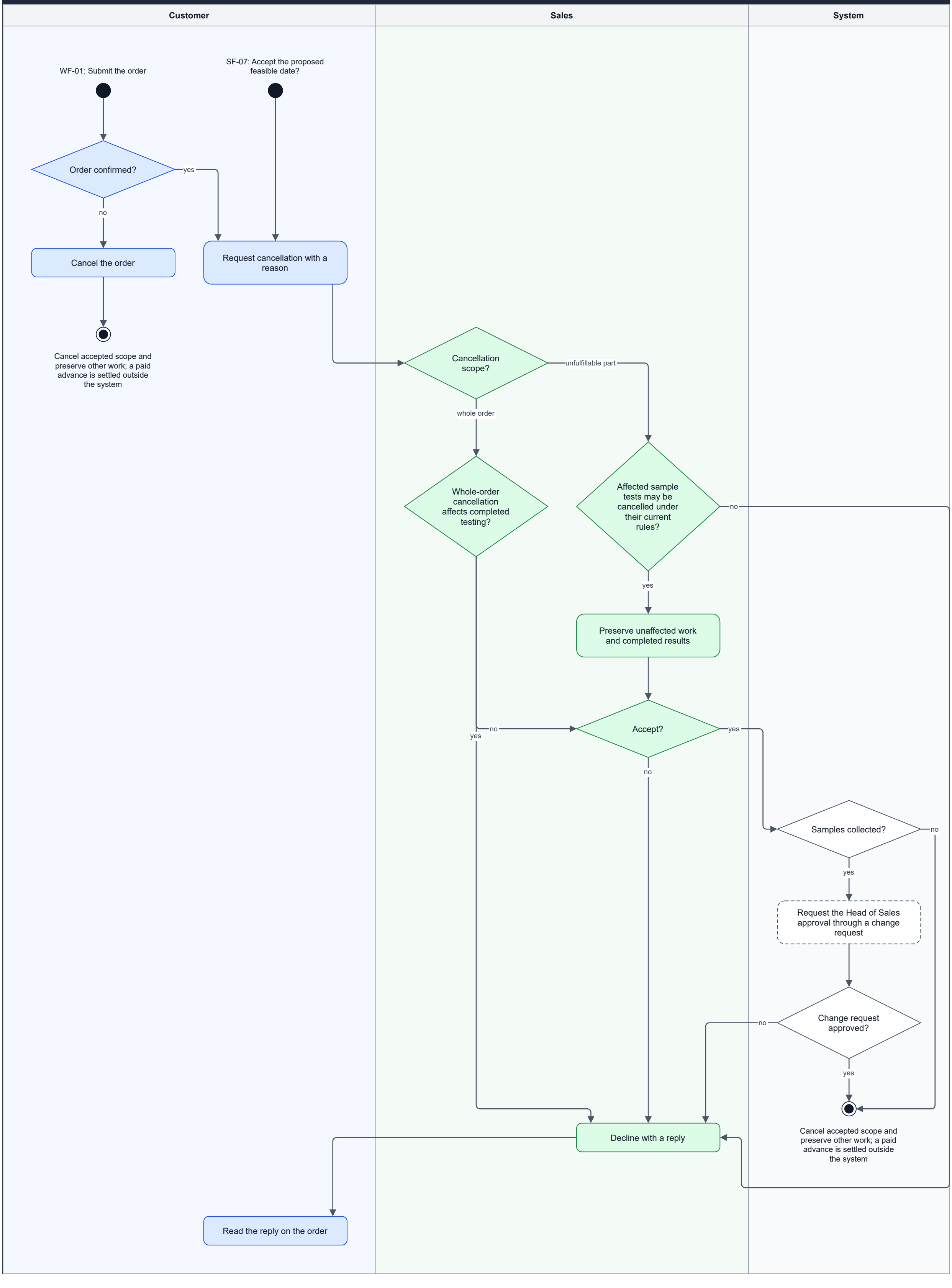




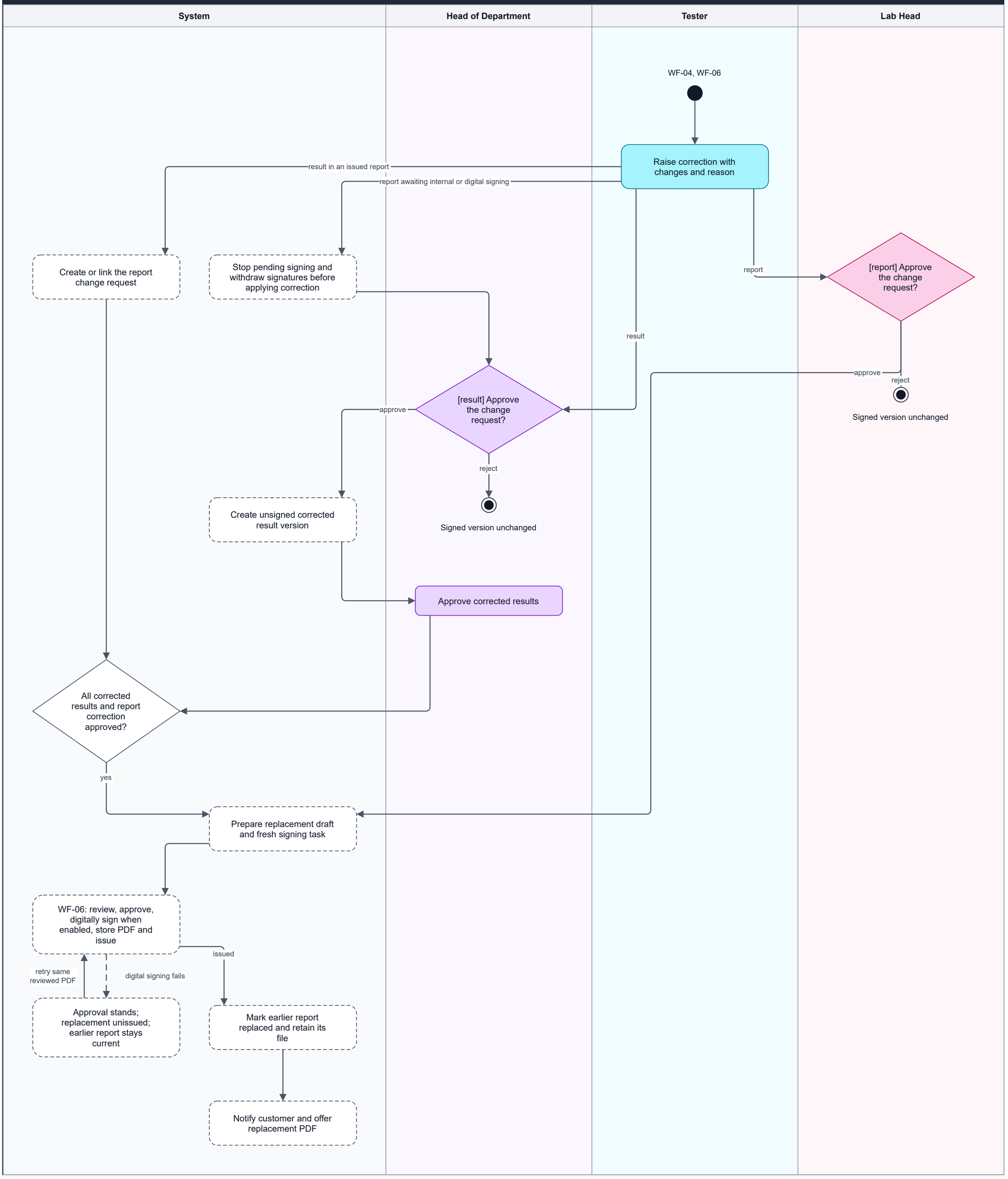
SF-01 Invoicing and payment collection

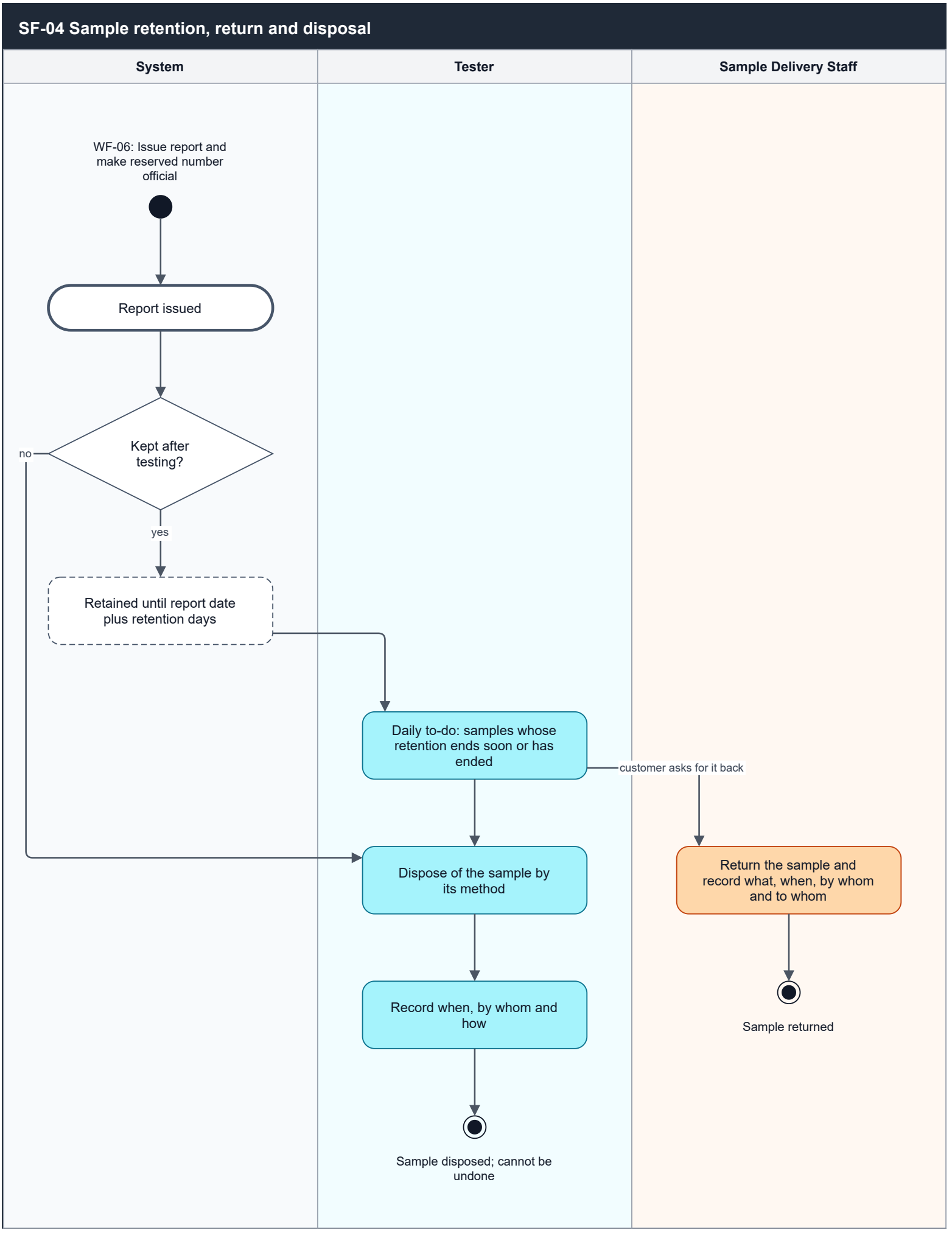


SF-02 Order cancellation

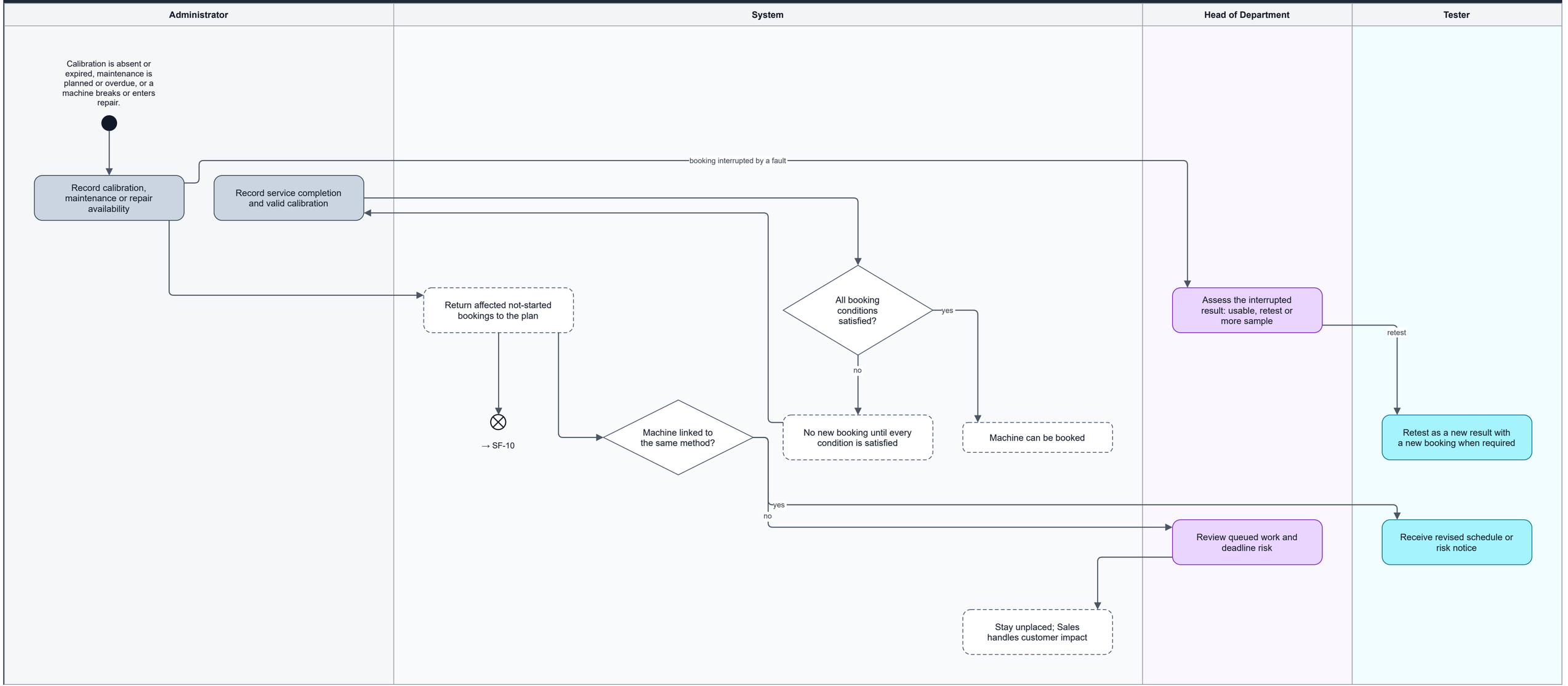


SF-03 Change request on a signed document

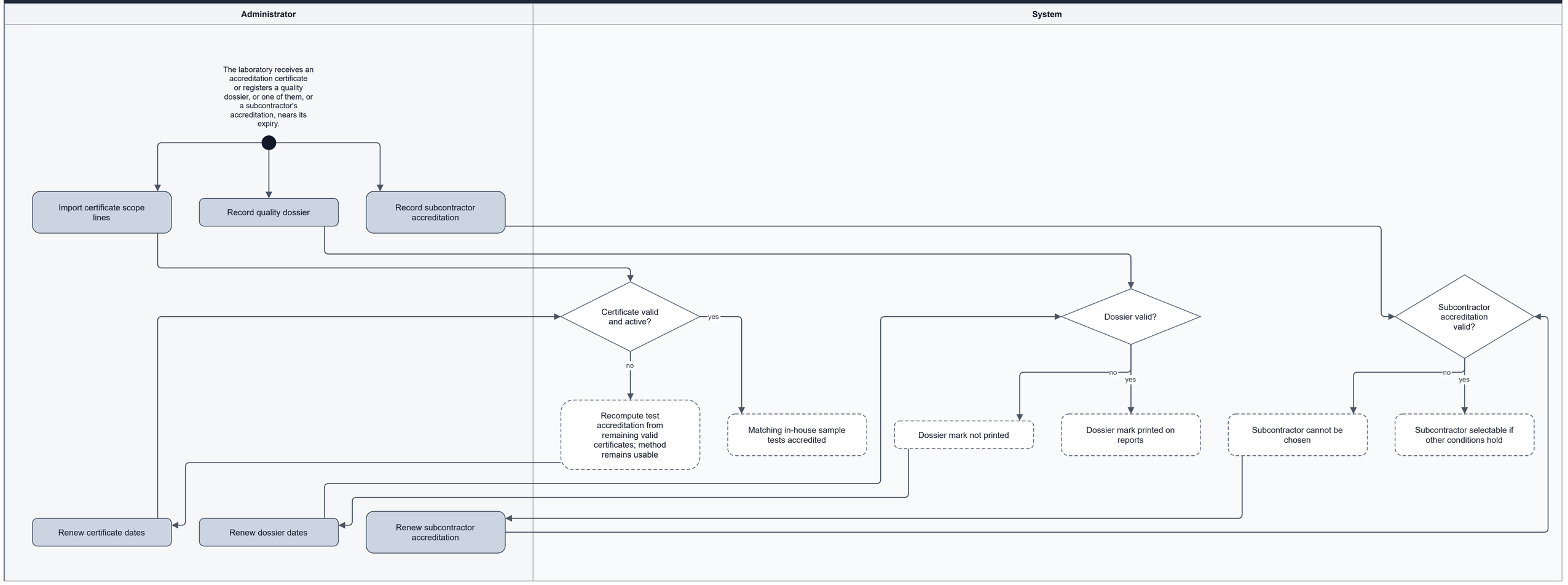



SF-04 Sample retention, return and disposal
System
Tester
Sample Delivery Staff
WF-06: Issue report and make reserved number official
Report issued
Kept after testing?
no
yes
Retained until report date plus retention days
Daily to-do: samples whose retention ends soon or has ended
customer asks for it back
Dispose of the sample by its method
Record when, by whom and how
Sample disposed; cannot be undone
Return the sample and record what, when, by whom and to whom
Sample returned

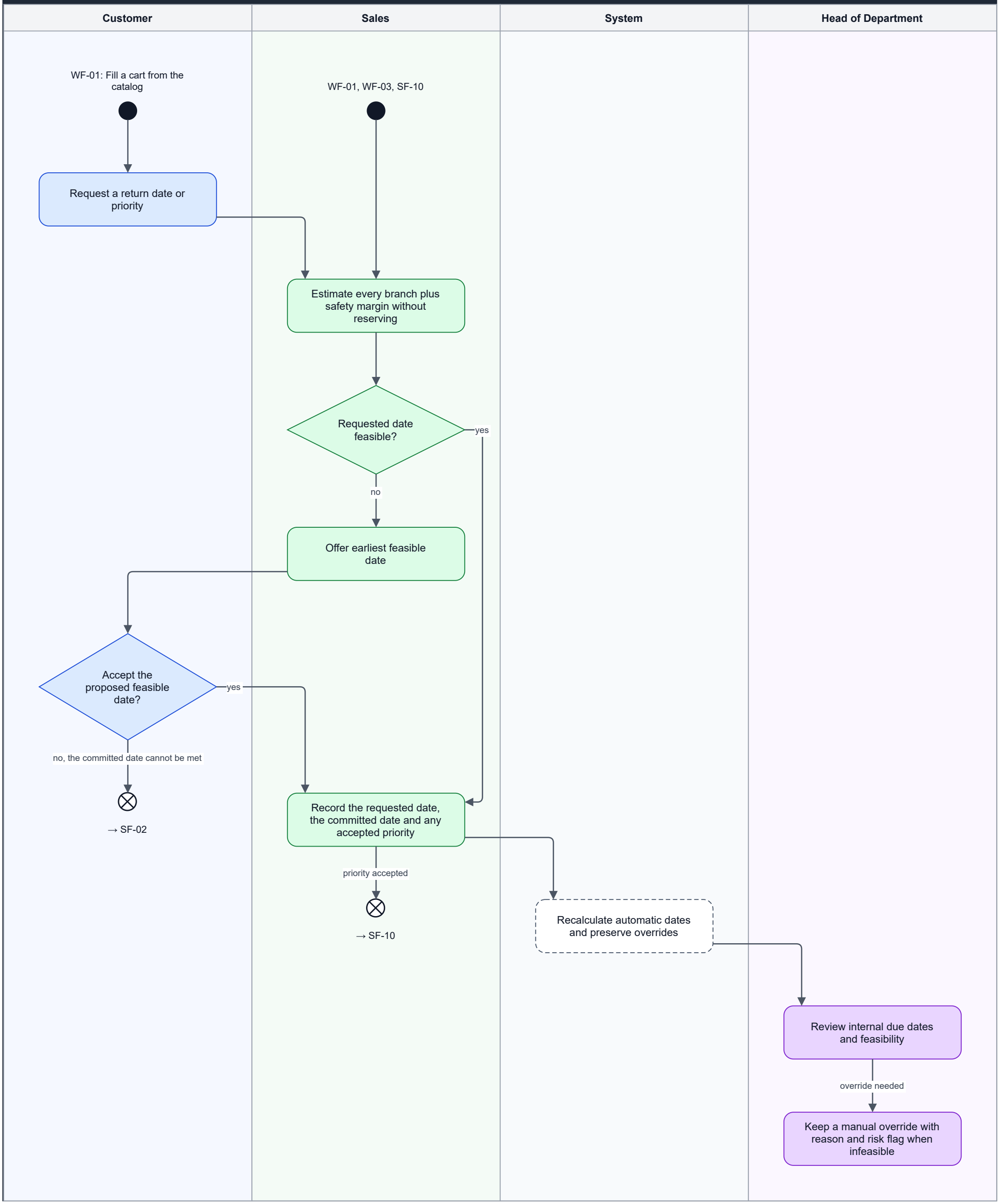
SF-05 Machine calibration and unavailability



SF-06 Accreditation and registration renewal



SF-07 Result dates and priority orders



SF-08 Laboratory setup

Administrator

A laboratory starts using MediLab, or its catalog, people or rules change.

Settings: deadlines, travel buffer and warning windows

Units, sample types and parameter groups

Report templates: edit, preview and make versions available

People, roles and approval chains

Test parameters

Ways and explicit type applicability: unit, LOD, LOQ and default

Optional regulations and limits per parameter and sample type

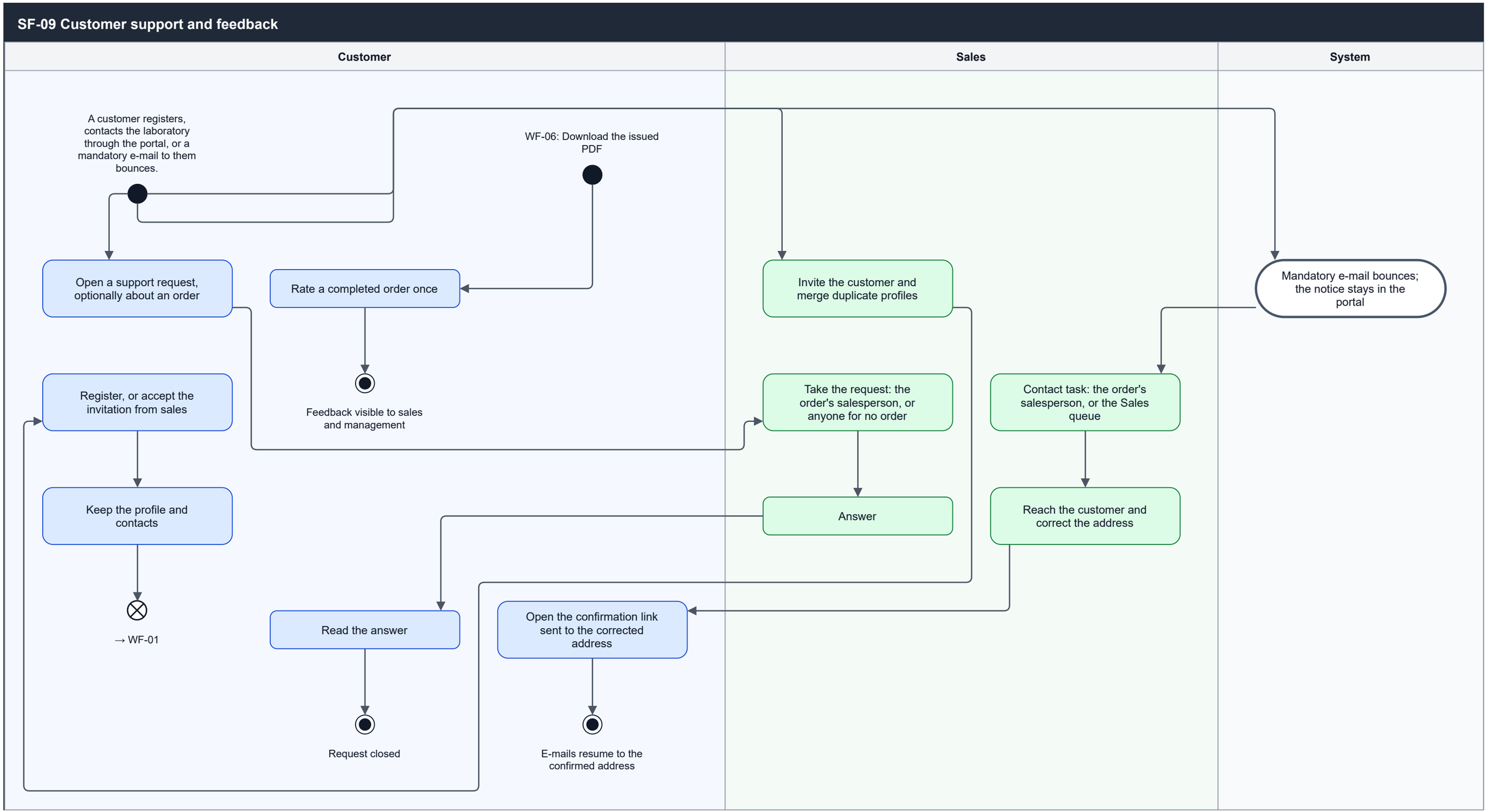
Sampling requirements and required resources

Prices, taxes and service packages

Lab Head

Authorize each tester's parameters without expiry

Ready for testing and reports → WF-01


SF-09 Customer support and feedback
Customer
Sales
System
A customer registers, contacts the laboratory through the portal, or a mandatory e-mail to them bounces.
WF-06: Download the issued PDF
Open a support request, optionally about an order
Rate a completed order once
Feedback visible to sales and management
Register, or accept the invitation from sales
Keep the profile and contacts
→ WF-01
Invite the customer and merge duplicate profiles
Take the request: the order's salesperson, or anyone for no order
Answer
Contact task: the order's salesperson, or the Sales queue
Reach the customer and correct the address
Mandatory e-mail bounces; the notice stays in the portal
Read the answer
Request closed
Open the confirmation link sent to the corrected address
E-mails resume to the confirmed address

SF-10 Scheduling plan and revisions

System

WF-03: Department queue

Received, not-started work

SF-05, SF-07

Update all reservations and reminders

Place by priority class, committed date and arrival

urgent work needs scheduled slots

Plan revision lists every moved booking

a move would miss a committed date

Hold it back; its booking keeps its slot

→ SF-07

Apply the moves; running bookings never move

Accept: reserve tester, required locations and machine if needed

deadline risk or resource disruption

Head of Department

Move or cancel a department booking, which pins it

Approve the moves of the department?

approve

Free capacity with OverTime or another machine

Assess risk, leave or unavailable room

→ WF-05

→ SF-07

Tester

Testing work arrives, or urgency, a booking change, a result, a cancellation, resource availability or a customer date changes the plan.

Move or cancel own not-started booking

Mark own test urgent with a reason

Start the booking

Enter the result and free the machine

Booking and result progress under their lifecycles

SF-11 Laboratory file administration

